Practical No 9

Aim: To develop test cases for any project using White-Box Testing.

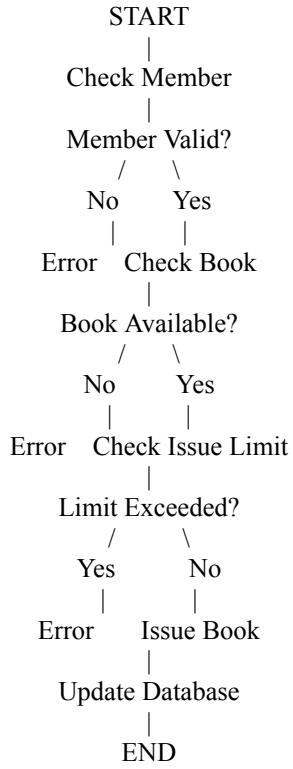
Objective:
To verify the internal logic, control flow, conditions, branches, and execution paths of the Library Management System.

Theory:
White-Box Testing is a software testing technique in which the internal structure and logic of the program are known and tested. Test cases are designed by examining the source code, control flow, conditions, loops, and paths.
Suppose for the Library Management System, white-box testing can be applied to modules such as:

- Login
- Member Management
- Book Search
- Book Issue
- Book Return
- Fine Calculation

Example: Book Issue Module

Consider the following logic:



White-Box Test Cases:

Test CaseID	Test Condition	Input	Expected Result
WB01	Valid member, available book, limit not exceeded	Valid Member ID, Available Book ID	Book is issued successfully
WB02	Invalid member	Invalid Member ID	"Invalid Member" message displayed
WB03	Valid member but book unavailable	Valid Member ID, Unavailable Book ID	"Book Not Available" message displayed

Test CaseID	Test Condition	Input	Expected Result
WB04	Valid member and book, but issue limit exceeded	Member with maximum books	"Issue Limit Exceeded" message displayed
WB05	Valid member and available book	Valid Member ID + Book ID	Issue record is created and database is updated

Fine Calculation Test Cases
Suppose:

```
If ReturnDate <= DueDate
    Fine = 0
Else
   DaysLate = ReturnDate - DueDate

  Fine = DaysLate × FinePerDay
```

Test Case ID	Condition	Due Date	Return Date	Expected Result
WB06	Returned before due date	10-Oct	8-Oct	Fine = ₹0
WB07	Returned on due date	10-Oct	10-Oct	Fine = ₹0
WB08	Returned after due date	10-Oct	11-Oct	Fine for 1 day
WB09	Returned after several days	10-Oct	15-Oct	Fine for 5 days

White-Box Testing Techniques

Technique	Application
Statement Coverage	Ensures every program statement is executed
Branch Coverage	Tests both True and False branches
Condition Coverage	Tests individual conditions
Path Coverage	Tests different execution paths
Loop Testing	Tests loops used for searching books, members, and records

Result:
The test cases for the Library Management System were successfully developed using White-Box Testing. The test cases verify the internal logic, statements, conditions, branches, and execution paths of the system.